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Answers & Explanations:

1. Answer: (B) Mitochondria

Explanation: Mitochondria generate most of the chemical energy needed to power the biochemical reactions.

2. Answer: (A) Diamond (Carbon)

Explanation: Diamond has thermal conductivity exceeding $2000 \text{ W/(m}\cdot\text{K)}$, significantly higher than silver.

3. Answer: True

Explanation: Sound speed in freshwater is approximately 1480 m/s compared to 343 m/s in air.

4. Answer: (A) $x = 3, 4$

Explanation: Factoring $(x - 3)(x - 4) = 0$ gives roots 3 and 4.

5. Answer: Review

Explanation: Long answer question graded by human evaluator.